Day 113 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Find Pyramid Surface Area

Standards: 6.G.A.4

Learning target: I can find pyramid surface area from its triangular faces and base.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. For a square pyramid, draw one square and four triangles. Each triangle has area base edge × face height / 2. The face height is measured on the triangular face, not vertically inside the solid.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4

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01 / Learn

For a square pyramid, draw one square and four triangles. Each triangle has area base edge × face height / 2. The face height is measured on the triangular face, not vertically inside the solid.

Worked example 1

A square pyramid has base edge 6 cm. Each triangular face has perpendicular face height 6 cm. Draw a net and calculate total surface area.

1. For a square pyramid, draw one square and four triangles.

2. Each triangle has area base edge × face height / 2.

3. Base = 36

4. each triangle = 6 × 6 ÷ 2 = 18

5. add four triangles to get 108.

6. State the result in the form requested: 108 cm².

Answer: 108 cm²

Worked example 2

A square pyramid net has base side 8 cm and four triangular faces, each with perpendicular face height 7 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

1. For a square pyramid, draw one square and four triangles.

2. One face = (8 × 7)/2 = 28.

3. Four faces total 112.

4. Add square base area 64.

Answer: Face 28 cm²; total 176 cm²

Day 113 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 8 cm. Draw a net and calculate total surface area.

02 / Constructed response

A square pyramid has base edge 10 cm. Each triangular face has perpendicular face height 12 cm. Draw a net and calculate total surface area.

03 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 13 cm. Draw a net and calculate total surface area.

04 / Constructed response

A square pyramid has base edge 4 cm. Each triangular face has perpendicular face height 6 cm. Draw a net and calculate total surface area.

05 / Constructed response

A square pyramid has base edge 8 cm. Each triangular face has perpendicular face height 8 cm. Draw a net and calculate total surface area.

06 / Constructed response

A square pyramid has base edge 8 cm. Each triangular face has perpendicular face height 7 cm. Draw a net and calculate total surface area.

07 / Constructed response

A square pyramid has base edge 6 cm. Each triangular face has perpendicular face height 7 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed.

08 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 8 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

09 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 10 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

10 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 11 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed.

Day 113 | Unit 08 | Answer key

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1. 336 cm²

Base = 144; each triangle = 12 × 8 ÷ 2 = 48; add four triangles to get 336.

6.G.A.4 | Constructed response

2. 340 cm²

Base = 100; each triangle = 10 × 12 ÷ 2 = 60; add four triangles to get 340.

6.G.A.4 | Constructed response

3. 456 cm²

Base = 144; each triangle = 12 × 13 ÷ 2 = 78; add four triangles to get 456.

6.G.A.4 | Constructed response

4. 64 cm²

Base = 16; each triangle = 4 × 6 ÷ 2 = 12; add four triangles to get 64.

6.G.A.4 | Constructed response

5. 192 cm²

Base = 64; each triangle = 8 × 8 ÷ 2 = 32; add four triangles to get 192.

6.G.A.4 | Constructed response

6. 176 cm²

Base = 64; each triangle = 8 × 7 ÷ 2 = 28; add four triangles to get 176.

6.G.A.4 | Constructed response

7. 120 cm²

Base = 36; each triangle = 6 × 7 ÷ 2 = 21; add four triangles to get 120.

6.G.A.4 | Constructed response

8. Face 48 cm²; total 336 cm²

One face = (12 × 8)/2 = 48. Four faces total 192. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

9. Face 60 cm²; total 384 cm²

One face = (12 × 10)/2 = 60. Four faces total 240. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

10. 408 cm²

Base = 144; each triangle = 12 × 11 ÷ 2 = 66; add four triangles to get 408.

6.G.A.4 | Constructed response